📊 E-Commerce Customer Churn, Cohort Retention & RFM Analysis

DOMAIN	BUSINESS ANALYTICS
LANGUAGE	SQL (POSTGRESQL)
ANALYTICS	PYTHON (PANDAS)
VISUALIZATION	POWER BI / EXCEL
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📌 Executive Summary

Customer acquisition costs (CAC) in modern e-commerce platforms have increased by over 45% in recent years, making **customer retention** and **repeat purchase frequency** paramount to profitability.

This project provides an end-to-end business analysis of customer churn across **1,200 transactions** totaling **₹1.10 Crore in gross revenue**. By implementing **Cohort Retention Analysis** and a **Recency-Frequency-Monetary (RFM) Segmentation Model**, this study uncovers the critical drivers of customer drop-off and formulates data-backed retention interventions for product and marketing stakeholders.

🎯 Business Problem Statement

- **High First-Order Drop-off:** Over **78% of customers** did not make a repeat purchase within 60 days following their initial order.
- **Lack of Customer Segmentation:** Marketing sent uniform broadcast discount emails rather than segmenting users by purchase recency and customer lifetime value (LTV).
- **Objective:**
  1. Measure true retention curves month-over-month using SQL cohort modeling.
  2. Segment 380 unique customer accounts into 4 actionable behavioral profiles.
  3. Formulate targeted business strategies to cut churn by 15% and protect high-LTV accounts.

🛠️ Analytics Tech Stack

- **Database & Querying:** SQL (PostgreSQL) — Common Table Expressions (`WITH`), Window Functions (`NTILE`, `LAG`, `ROW_NUMBER`), Multi-table Joins, Cohort Grouping.
- **Data Processing & Modeling:** Python (Pandas, NumPy) for RFM quantile scoring and summary distributions.
- **Business Documentation:** Business Requirement Document (BRD) & Functional Specifications.
- **Visualization:** Power BI & MS Excel (Pivot Tables, Dynamic KPI Cards).

📈 Key Findings & Business Insights

1. RFM Behavioral Segmentation Breakdown

Customer Segment	% of Customer Base	Characteristics	Business Strategy
Champions (High Loyalty)	**32.4%**	Highly recent, high order frequency & value	VIP loyalty rewards, early product drops, no heavy discounts needed.
Promising (New Shoppers)	**18.2%**	Recent first purchases, low frequency	Onboarding drip emails, category discovery vouchers within Day 7–14.
At Risk (High LTV / Inactive)	**17.6%**	Historically high spenders, inactive >60 days	Re-engagement SMS/WhatsApp with personalized "We Miss You" perks.
Hibernating / Churned	**31.8%**	Long period of inactivity, low repeat rate	Win-back campaign with clearance promos; cap paid ad re-targeting to save budget.

2. Category Performance & Churn Correlation

- **Books & Beauty:** Exhibited the highest churn rates (~80.5% – 83.3%), despite strong gross revenue. Customers often buy single gift or seasonal items without repurchasing.
- **Apparel & Home/Kitchen:** Showed the highest repeat retention (~23.8% repeat buyers), making them the primary anchor categories for subscription/loyalty bundling.

💡 Strategic Recommendations for Stakeholders

1. **Automate Post-Purchase Trigger Workflows:** Trigger a personalized product recommendation email on **Day 10 post-delivery** (before the 60-day churn threshold is reached).
2. **Re-allocate Marketing Budget:** Divert 25% of top-of-funnel acquisition ad spend toward retention campaigns targeting the **17.6% "At Risk" segment**, preventing high-value revenue leakage.
3. **Loyalty & Gamification:** Introduce tiered loyalty points for repeat orders within 30 days to incentivize faster purchase cycles.

📂 Repository File Structure

```
├── customer_churn_analysis.sql    # Complete SQL script (KPIs, Cohorts, RFM)
├── churn_analysis.py              # Automated Python data modeling pipeline
├── sample_ecommerce_data.csv      # Raw transaction dataset (1,200 records)
├── Executive_Project_Report.pdf   # 2-Page Executive Presentation for Stakeholders
└── README.md                      # Comprehensive project documentation
```

🚀 How to Run the Analysis Locally

1. Clone the repository:
   `bash git clone https://github.com/Nakshatra2016/Ecommerce-Customer-Churn-Retention-Analysis.git cd Ecommerce-Customer-Churn-Retention-Analysis`
2. Run the SQL script against any PostgreSQL or MySQL client:
   `bash psql -U your_user -d your_db -f customer_churn_analysis.sql`
3. Or run the automated Python pipeline:
   `bash python3 churn_analysis.py`

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